Loop integrals in de Sitter spacetime: The parity-split IBP system and d log-form differential equations

Jiaqi Chen[1,2],* Bo Feng[3,4],† Zhehan Qin[5],‡ and Yi-Xiao Tao[6]§
[1] *Beijing Key Laboratory of Optical Detection Technology for Oil and Gas, China University of Petroleum-Beijing, Beijing 102249, China*
[2] *Basic Research Center for Energy Interdisciplinary, College of Science, China University of Petroleum-Beijing, Beijing 102249, China*
[3] *State Key Laboratory of Nuclear Physics and Technology, Institute of Quantum Matter, South China Normal University, Guangzhou 510006, China*
[4] *Guangdong Basic Research Center of Excellence for Structure and Fundamental Interactions of Matter, Guangdong Provincial Key Laboratory of Nuclear Science, Guangzhou 510006, China*
[5] *Department of Physics, Tsinghua University, Beijing 100084, China*
[6] *Department of Mathematical Sciences, Tsinghua University, Beijing 100084, China*

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We develop integration-by-parts (IBP) reduction and differential equations for massive loop integrals of cosmological correlators in de Sitter (dS) spacetime, demonstrating the feasibility of this approach. We identify a structural property of the dS IBP system: for an n-propagator family, it splits into 2^n closed subsystems classified by the parity of the propagator indices. We further formulate a Baikov representation for loop integrals in dS space and derive the corresponding dimensional recurrence relations. In flat spacetime, intersection theory shows that d log-form master integrands lead to d log-form differential equations. Motivated by fibration intersection theory, we conjecture that this construction extends to dS integrands involving Hankel functions. We verify this conjecture in the one-loop bubble family and determine the associated alphabet.

Keywords: Cosmology, Scattering amplitudes, Quantum field theory in curved spacetime, Feynman integrals

I. INTRODUCTION

De Sitter (dS) spacetime is of central phenomenological importance in inflationary cosmology [1] and at the same time provides the simplest curved background for developing systematic perturbative methods. For massless or conformally coupled fields, the integrands simplify considerably and often retain structures closely analogous to the polynomial-type integrands familiar from flat space [2–20]. This has led both to amplitude-inspired developments [21–29] and to direct extensions of flat-space techniques such as integration-by-parts (IBP) reduction [30] and differential equations [31–34]. For correlators with massive intermediate states, several powerful complementary analytic approaches have also been proposed, including the cosmological bootstrap [35–50], Mellin techniques [44, 51–58], spectral decompositions [49, 59–64] and dispersive methods [65, 66]. See also [67–71] for some recent numerical studies.

Nevertheless, explicit computations of such correlators, which are central to cosmological collider physics [72–76], remain largely limited to tree level and to bubble topologies at one loop. Among currently available analytic tools, only spectral decompositions [60, 63, 77] and the partial Mellin-Barnes representation [58] have been successfully applied to one-loop bubble diagrams, while extending them to other important one-loop topologies, such as triangles and boxes, appears less straightforward. Existing results beyond bubbles include triangles of conformally coupled or massless scalars [17], a special case in which the triangle arises from a free theory via field redefinitions [78], and partial results for triangle and box diagrams obtained from partial Mellin-Barnes representations [79]. By contrast, IBP reduction is organized at the level of integral families and is therefore naturally suited for systematic generalization.

From the technical viewpoint, the massive case is substantially more challenging because the relevant integrands involve Hankel functions and are no longer of polynomial type. Nevertheless, Ref. [80] pointed out that IBP reduction and the method of differential equations can still be extended to massive dS correlators. At tree level, and for the time-integral part at loop level, this leads to differential equations in d log-form [71, 81, 82] and new analytic solutions in terms of hypergeometric functions [71].

In this letter, we take a first step toward the complete loop-level massive case, including both the time and loop-momentum integrals. We formulate the IBP reduction for this family and show that the reduction system decomposes into independent sectors labeled by the odd/even parity of the indices associated with each propagator. More generally, for an n-propagator family, this implies a decomposition into 2^n IBP-closed subsystems, greatly simplifying the reduction of dS loop integrals.

We also investigate the analytic structure of the associated differential equations. In flat spacetime, canonical differential equations [83] in d log-form play a central role in the analytic computation of Feynman integrals.

* jiaqichen@cup.edu.cn
† fengbo@scnu.edu.cn
‡ qzh21@mails.tsinghua.edu.cn
§ taoyx21@mails.tsinghua.edu.cn

It has long been observed that d log-form (or leading-singularity) master integrands [84–90] lead to d log-form differential equations, and intersection theory [87, 88, 91–108] provided a natural framework for understanding this mechanism [109, 110]. Motivated by [109, 110] and the fibration approach to multivariate intersection numbers [92], we propose a generalized perspective adapted to the massive dS case, in which the master integrals of the τ-integrated kernel are treated as a twist $\mathbf{U}$, while the remaining part of the integrand is encoded in a differential form Φ. This leads to a practical criterion for obtaining d log-form differential equations through an a priori construction of the integrand.

Using a dS version of the Baikov representation [111], we implement this construction for the one-loop bubble family, verify that the homogeneous top-sector block of the differential-equation system is in d log-form, and determine the corresponding alphabet explicitly. Taken together, these results provide evidence that loop-level massive cosmological correlators in dS space admit a systematic treatment that parallels the flat-space case despite the non-polynomial structure of the integrands.

II. SETUP

We work in the Poincaré patch of de Sitter space, with conformal time $\tau \in (-\infty, 0)$, where the metric takes the form

$$\mathrm{d}s^2 = a^2(\tau)\big(-\mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2\big)\,, \qquad a(\tau) = -\frac{1}{H\tau}\,. \tag{1}$$

For convenience, we set $H = 1$ throughout. Although most of our discussion applies to general spatial dimension d, the cosmologically relevant case is $d = 3$; dimensional regularization is implemented by analytic continuation to $d = 3 - 2\epsilon$. Following Ref. [80], we introduce the rescaled Hankel building blocks [112]

$$h_\nu^{(\alpha)}(z) \equiv z^{-\nu}\mathrm{H}_\nu^{(\alpha)}(z)\,, \qquad \alpha = 1, 2\,, \quad \nu \in \mathbb{C}\,, \tag{2}$$

which satisfy the rescaled Bessel equation

$$\partial_z^2 h_\nu^{(\alpha)}(z) + \frac{2\nu+1}{z}\partial_z h_\nu^{(\alpha)}(z) + h_\nu^{(\alpha)}(z) = 0\,. \tag{3}$$

When the label ν is clear from the context, we suppress it and denote derivatives by

$$\partial_z^n h_\nu^{(\alpha)}(z) \equiv h_n^{(\alpha)}(z)\,, \tag{4}$$

Once h_2 appears, we will reduce it to h_0 and h_1 immediately by (3). We use the standard Schwinger-Keldysh (SK) formalism [113–116] to compute equal-time correlators, following the diagrammatic rules of Ref. [117]. As a first example, we consider the 1-loop bubble diagram with four external legs of the conformally coupled scalar ϕ_c (while the minimally coupled case shares the same function family) mediated by a loop of massive scalars σ, as shown in Fig. 1.

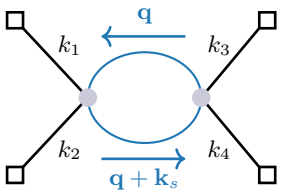

FIG. 1. SK diagrams for four-point correlators with bubble topology. The external lines represent conformally coupled scalars, while the blue loop lines represent massive scalars. The □ vertices lie on the future boundary, and the • vertices denote bulk insertions on the two Schwinger-Keldysh contours, with + and − corresponding to the time-ordered and anti-time-ordered branches, respectively. Contributions from all SK branches are summed.

The main technical difficulty lies in the bulk-to-bulk propagators,

$$\begin{aligned} G_{\pm\pm}(k;\tau_1,\tau_2) &= \theta_{12}G_{\mp\pm}(k;\tau_1,\tau_2) + \theta_{21}G_{\pm\mp}(k;\tau_1,\tau_2)\,,\\ G_{-+}(k;\tau_1,\tau_2) &\equiv u(k,\tau_1)u^*(k,\tau_2)\,,\\ G_{+-}(k;\tau_1,\tau_2) &\equiv u^*(k,\tau_1)u(k,\tau_2)\,, \end{aligned} \tag{5}$$

where the mode function and its complex conjugate are

$$\begin{aligned} u(k,\tau) &= \frac{\sqrt{\pi}}{2}e^{\mathrm{i}\pi\nu/2}(-\tau)^{d/2}\mathrm{H}_\nu^{(1)}(-k\tau)\,,\\ u^*(k,\tau) &= \frac{\sqrt{\pi}}{2}e^{-\mathrm{i}\pi\nu^*/2}(-\tau)^{d/2}\mathrm{H}_{\nu^*}^{(2)}(-k\tau)\,. \end{aligned} \tag{6}$$

Here $\nu \equiv \sqrt{d^2/4 - m^2}$ is the mass parameter. For notational simplicity, we focus on the complementary-series case where ν is real, while our method applies equally well to the principal-series case where ν is purely imaginary. We also use the shorthand $\theta_{ij} \equiv \theta(\tau_i - \tau_j)$. By contrast, the bulk-to-boundary propagators of ϕ_c take the simple form

$$K_\pm(k;\tau) = \frac{\tau}{2k}e^{\pm\mathrm{i}k\tau}\,. \tag{7}$$

We denote by $\mathcal{T}_{\mathsf{ab}}$, with $\mathsf{a},\mathsf{b} \in \{+,-\}$, the contribution from the SK branch in which the vertices at τ_1 and τ_2 lie on the a and b contours, respectively. Of the four branches, $\mathcal{T}_{++}$ and $\mathcal{T}_{--}$ are more involved because G_{++} and G_{--} each contain both θ_{12} and θ_{21} terms. Since $\mathcal{T}_{++} = \mathcal{T}_{--}^*$, it suffices to consider $\mathcal{T}_{--}$. For the simplest contact interaction $a^{d+1}\phi_\mathrm{c}^2\sigma^2$, one finds

$$\begin{aligned} \mathcal{T}_{--} \propto \int_{-\infty}^{0} &\frac{\mathrm{d}\tau_1}{(-\tau_1)^{d+1}}\frac{\mathrm{d}\tau_2}{(-\tau_2)^{d+1}}\\ &\times K_-(k_1;\tau_1)K_-(k_2;\tau_1)K_-(k_3;\tau_2)K_-(k_4;\tau_2)\\ &\times \int\frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\,G_{--}(|\mathbf{q}|;\tau_1,\tau_2)G_{--}(|\mathbf{q}+\mathbf{k}_s|;\tau_1,\tau_2)\,. \end{aligned} \tag{8}$$

Here k_i is the magnitude of $\mathbf{k}_i$. This motivates the following integral family for the bubble topology, which accommodates vertices with arbitrary numbers of derivatives:

$$I[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] = \int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\frac{1}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} I^{\tau}_{\{\boldsymbol{n},a_1,a_2\}}\,, \qquad I^{\tau}_{\{\boldsymbol{n},a_1,a_2\}} \equiv \int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\,\hat{I}^{\tau}_{\{\boldsymbol{n},a_1,a_2\}}\,,$$

$$\hat{I}^{\tau}_{\{\boldsymbol{n},a_1,a_2\}} = e^{-\mathrm{i}P_1\tau_1-\mathrm{i}P_2\tau_2}\,(-\tau_1)^{a_1}(-\tau_2)^{a_2}\left[\theta_{21}h^{(1)}_{n_1}(-|\mathbf{q}|\tau_1)h^{(2)}_{n_2}(-|\mathbf{q}|\tau_2)+\theta_{12}h^{(2)}_{n_1}(-|\mathbf{q}|\tau_1)h^{(1)}_{n_2}(-|\mathbf{q}|\tau_2)\right]$$
$$\times\left[\theta_{21}h^{(1)}_{n_3}(-|\mathbf{q}+\mathbf{k}_s|\tau_1)h^{(2)}_{n_4}(-|\mathbf{q}+\mathbf{k}_s|\tau_2)+\theta_{12}h^{(2)}_{n_3}(-|\mathbf{q}+\mathbf{k}_s|\tau_1)h^{(1)}_{n_4}(-|\mathbf{q}+\mathbf{k}_s|\tau_2)\right].$$

Here $P_1 = k_1 + k_2$ and $P_2 = k_3 + k_4$ are the energies flowing into the two vertices. Equation (3) implies that any $h^{(\alpha)}_n(z)$ with $n \geq 2$ can be recursively reduced to a linear combination of $h^{(\alpha)}_0(z)$ and $h^{(\alpha)}_1(z)$. It is thus sufficient to restrict all indices n_i to $\{0, 1\}$. The contribution in (8) then corresponds, up to an overall normalization, to

$$\mathcal{T}_{--} \propto I[\{0,0,0,0\},\{1+2\nu,1+2\nu\},\{-2\nu,-2\nu\}]\,. \quad (9)$$

The family (9) is manifestly closed under IBP identities generated by total derivatives in the loop momentum $\mathbf{q}$. Temporal IBP is more subtle, because derivatives acting on θ_{21} and θ_{12} generate contact terms proportional to $\delta(\tau_1 - \tau_2)$. As explained in Ref. [80], after using this delta function to localize one of the time integrals, the collapsed propagator is evaluated through the Wronskian relation for real ν:

$$h^{(1)}_{n_1}(z)\,h^{(2)}_{n_2}(z) - h^{(2)}_{n_1}(z)\,h^{(1)}_{n_2}(z) = (n_1 - n_2)\frac{4\mathrm{i}}{\pi}z^{-2\nu-1}\,, \quad (10)$$

with $n_{1,2} \in \{0, 1\}$. It follows that, under dτ-IBP, the family (9) closes up to residual terms belonging to the tadpole-like family

$$\begin{aligned} &R[\{n_1,n_2\},\{a\},\{b_1,b_2\}] \\ &\equiv \frac{4\mathrm{i}}{\pi}\int_{-\infty}^{0}\mathrm{d}\tau\int\frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\frac{e^{-\mathrm{i}P_0\tau}(-\tau)^a}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} \\ &\times\frac{1}{2}\sum_{i=0,1}h^{(1+i)}_{n_1}(-|\mathbf{q}|\tau)h^{(2-i)}_{n_2}(-|\mathbf{q}|\tau) \end{aligned} \quad (11)$$

where $P_0 \equiv P_1 + P_2$. When $b_2 = 0$, R reduces to the standard tadpole family.

For comparison, $\mathcal{T}_{-+}$, in which the first bulk vertex lies on the + contour and the second on the − contour, is naturally encoded in the family

$$\begin{aligned} \widetilde{I}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] &= \int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\int\frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d} \\ &\times e^{-\mathrm{i}P_1\tau_1+\mathrm{i}P_2\tau_2}\,\frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}}h^{(1)}_{n_1}(-|\mathbf{q}|\tau_1)h^{(2)}_{n_2}(-|\mathbf{q}|\tau_2) \\ &\times h^{(1)}_{n_3}(-|\mathbf{q}+\mathbf{k}_s|\tau_1)h^{(2)}_{n_4}(-|\mathbf{q}+\mathbf{k}_s|\tau_2)\,. \end{aligned} \quad (12)$$

Similar to (9), we have

$$\mathcal{T}_{-+} \propto \widetilde{I}[\{0,0,0,0\},\{1+2\nu,1+2\nu\},\{-2\nu,-2\nu\}]\,. \quad (13)$$

Since (12) contains no step functions, it closes under both dq-IBP and dτ-IBP, with no additional tadpole-like family. Its IBP relations follow from those of (9) upon replacing $P_2 \to -P_2$ and discarding all terms proportional to R. We therefore restrict attention to the family (9). The complete IBP relations are given in the Supplemental Material. The top-sector relations of (9) immediately reveal the following conserved *parities*:

1. For both dq-IBP and dτ-IBP, the parities of $n_1 + n_2 + b_1$ and $n_3 + n_4 + b_2$ are conserved. Each conserved parity is naturally associated with one propagator.

2. For dq-IBP, the parities of $n_1 + n_3 + a_1$ and $n_2 + n_4 + a_2$ are also conserved [118].

In particular, in the residual family R defined in (11), the line carrying momentum $\mathbf{k}_s + \mathbf{q}$ no longer contains any Hankel functions, so the corresponding conserved parity is simply the parity of b_2.

More generally, the first conservation pattern extends straightforwardly to n-propagator families, yielding 2^n IBP-closed subsystems and thus significantly simplifying the reduction of dS loop integrals. For simplicity, in the following part, we restrict attention to the even-even subsystem with $P_1 = P_2$:

$$n_1 + n_2 + b_1 \text{ even}\,, \qquad n_3 + n_4 + b_2 \text{ even}\,. \quad (14)$$

We reduce this subsystem using the user-defined system module of Kira [119–121]. It contains 14 top-sector master integrals and 5 in the remaining-term family. The five remaining-term masters are genuine members of the closed even-even system: temporal-IBP contact terms allow derivatives of top-sector masters to contain them. The construction below concerns the homogeneous top-to-top block of this complete system.

III. d log INTEGRANDS OF THE TOP SECTOR AND THE ALPHABET

In this section, we construct the d log-form integrands for the top sector and determine the alphabet of the associated differential system. Since the Baikov representation is particularly well suited to the construction of d log-forms in flat spacetime, we first adapt it to cosmological correlators and then explain how a viewpoint inspired by fibration intersection theory motivates our construction.

For the dS bubble diagram, we introduce the Baikov variables [111]

$$x_1 = \sqrt{z_1} = |\mathbf{q}|\,, \qquad x_2 = \sqrt{z_2} = |\mathbf{q}+\mathbf{k}_s|\,. \tag{15}$$

Using the same change of variables as in the flat-space Baikov representation, one finds the normalized measure

$$\frac{2}{\pi^{d/2}}\,\mathrm{d}^d q = C_d\,\mathcal{G}^{\frac{d-3}{2}}\mathcal{K}^{-\frac{d-2}{2}}\,\mathrm{d}z_1\,\mathrm{d}z_2\,,\ C_d = \frac{\pi^{-1/2}}{\Gamma[(d-1)/2]}\,,$$
$$\mathcal{K} = k_s^2\,,\ \mathcal{G} = \frac{1}{4}\Big(2z_1k_s^2 + 2z_2k_s^2 - k_s^4 - z_1^2 - z_2^2 + 2z_1z_2\Big)\,, \tag{16}$$

valid for general spatial dimension d. For later use, note that $\mathrm{d}z_i = 2x_i\,\mathrm{d}x_i$, so that a factor $x_i^{-b_i}$ in the z_i representation is shifted by one power when rewritten in terms of x_i:

$$I[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] \sim \int\cdots\times\frac{\mathrm{d}z_1}{x_1^{b_1}}\frac{\mathrm{d}z_2}{x_2^{b_2}}$$
$$= \int\cdots\times 4\,\frac{\mathrm{d}x_1}{x_1^{b_1-1}}\frac{\mathrm{d}x_2}{x_2^{b_2-1}}\,. \tag{17}$$

The Baikov representation also allows the dimensional recurrence relations [122, 123] to be carried over from flat spacetime to dS space; the relevant relation is given in (67).

Since IBP shifts the indices a_i and b_i only by integers, it is convenient to absorb the common ν-dependent offsets into the index notation and rewrite

$$I[\{\boldsymbol{n}\},\{a_0+a_1',a_0+a_2'\},\{b_0+b_1',b_0+b_2'\}]$$
$$\to I[\{\boldsymbol{n}\},\{a_1',a_2'\},\{b_1',b_2'\}] \tag{18}$$

where, as seen from (9), these shifts are

$$a_0 = 2\nu\,, \quad b_0 = -2\nu\,. \tag{19}$$

We now briefly recall the relevant aspect of intersection theory. In flat spacetime, a Baikov integrand can be written as $u\,\phi$, with $u = \prod_i P_i^{\beta_i}$ and ϕ a d log form, for example $u = \mathcal{G}^{-\epsilon}$, so that the integrand is built entirely from polynomial factors. The associated twisted connection,

$$\omega \equiv \frac{\mathrm{d}u}{u} = \sum_i \beta_i\,\mathrm{d}\log P_i\,, \tag{20}$$

is therefore naturally of d log-form. It was further shown in [109, 110] that d log-form master integrands $\{\phi_i\}$ with d log-form ω lead to d log-form differential equations.

Intersection theory is usually formulated for polynomial-type integrands, whereas the dS integrands relevant here involve Hankel functions. To address this, we reinterpret the intermediate step in the fibration approach to intersection theory [92] as suggesting a generalized intersection-theoretic structure acting on objects of the form $\Phi\cdot\mathbf{U}$. Here $\mathbf{U}$ denotes a vector of master integrals for a kernel family, while Φ collects the remaining part of the integrand. Unlike in the standard setting, we do not require the integrand defining $\mathbf{U}$ to be of polynomial type.

For dS loop integrals, the τ integral can be incorporated into the kernel $\mathbf{U}$. Unlike in flat spacetime, the twisted connection Ω, defined by

$$\mathrm{d}\mathbf{U} = \Omega\cdot\mathbf{U}\,, \tag{21}$$

encodes the differential equations satisfied by the kernel integrals and is not automatically d log form. We conjecture that, if $\mathbf{U}$ is chosen such that Ω is in d log form and the remaining part of the integrands $\{\Phi\}$ are also chosen to be in d log form, then the corresponding differential equations are automatically in d log-form. We implement this idea below.

We choose the kernel to be

$$\mathbf{U}_{\boldsymbol{n}} = x_1^{-b_0}x_2^{-b_0}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\,I^{\tau}_{\boldsymbol{n}}\,, \tag{22}$$

where ϵ is the dimensional regulator and $d = 3-2\epsilon$. The one-loop Baikov measure contributes the factor $\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}$, and $I^{\tau}_{\boldsymbol{n}}$, with all n_i taking values 0 or 1, labels the master integrals of the τ-integral family, as studied in [80].

Ref. [80] also shows that the differential equations for $I^{\tau}_{\boldsymbol{n}}$ are in d log-form

$$\mathrm{d}I^{\tau}_{\boldsymbol{m}} = (\mathrm{d}\Omega^{\tau}_{\boldsymbol{mn}})\,I^{\tau}_{\boldsymbol{n}}\,, \quad \mathrm{d}\Omega^{\tau} = \mathrm{d}\Omega_{x_1} + \mathrm{d}\Omega_{x_2} + \mathrm{d}\Omega_{ex}\,, \tag{23}$$

where the one-forms in $\mathrm{d}\Omega^{\tau}$ are built from $\mathrm{d}\log x_1$, $\mathrm{d}\log x_2$, and $\mathrm{d}\log(P_i \pm x_1 \pm x_2)$, and the three contributions are denoted by $\mathrm{d}\Omega_{x_1}$, $\mathrm{d}\Omega_{x_2}$, and $\mathrm{d}\Omega_{ex}$, respectively. Accordingly, the twisted connection associated with $\mathbf{U}$ is also of d log form,

$$\mathrm{d}\mathbf{U} = \Omega\cdot\mathbf{U}$$
$$= \Big(\mathrm{d}\Omega^{\tau} - b_0(\mathrm{d}\log x_1 + \mathrm{d}\log x_2)$$
$$- \epsilon(\mathrm{d}\log\mathcal{G} - \mathrm{d}\log\mathcal{K})\Big)\cdot\mathbf{U}\,. \tag{24}$$

For fixed z_2, the two zeros of $\mathcal{G}$, viewed as a polynomial in z_1, are $z_1 = (k_s \pm \sqrt{z_2})^2$, so we have the following useful building blocks:

$$\mathrm{d}\log z_1 = 2\mathrm{d}\log x_1\,,$$
$$\mathrm{d}\log\left(\frac{(k_s+\sqrt{z_2})^2 - z_1}{(k_s-\sqrt{z_2})^2 - z_1}\right) = -\frac{2k_sx_1x_2\,\mathrm{d}x_1}{\mathcal{G}}$$
$$- \frac{k_s^3 - k_sx_1^2 - k_sx_2^2}{\mathcal{G}}\mathrm{d}x_2 \tag{25}$$

and $(z_1 \leftrightarrow z_2)$. The connection $\mathrm{d}\Omega^{\tau}$ contains additional d log-type building blocks. However, some of them involve denominators of the form $P_i \pm x_1 \pm x_2$, and using such factors directly would enlarge the integral family. To avoid this, we only use combinations that can be rewritten back in terms of the original family:

$$\partial_{P_j}\Omega_{ex}\,\mathrm{d}x_i = \partial_{P_j}\Omega^{\tau}\,\mathrm{d}x_i\,,$$
$$\partial_{x_j}\Omega_{ex}\,\mathrm{d}x_i = (\partial_{x_j}\Omega^{\tau} - \partial_{x_j}\Omega_{x_j})\,\mathrm{d}x_i\,, \tag{26}$$

When acting on $I^{\tau}_{\boldsymbol{n}}$, the first relation amounts, up to an overall constant, to a τ_j insertion. The second can be mapped back to the original family by an x_j-IBP, while $\partial_{x_j}\Omega_{x_j}$ contains only $1/x_j$ and therefore introduces no new denominator structure. Further details are given in the Supplemental Material.

Pairing these x_1- and x_2-fiber one-forms, we obtain the following five classes of logarithmic two-forms. Their representatives in the original family are obtained from $\int \Phi_{\boldsymbol{mn}}\mathbf{U}_{\boldsymbol{n}}$ with $\mathbf{U}$ given in (22).

(a) The simplest scalar pairing combines the Gram logarithmic one-form in (25) with $\mathrm{d}\log x_2$, namely, $\Phi_{\boldsymbol{mn}} \propto \delta_{\boldsymbol{mn}} \frac{x_1 x_2 k_s\,\mathrm{d}x_1}{\mathcal{G}} \frac{\mathrm{d}x_2}{x_2}$. This gives:

$$\begin{aligned} &I[\{0,0,0,1\},\{0,0\},\{0,1\},d=1-2\epsilon]\,,\\ &I[\{1,1,0,1\},\{0,0\},\{0,1\},d=1-2\epsilon]\,. \end{aligned} \tag{27}$$

(b) The kernel pairing combines the complete Gram logarithmic one-form in (25) with the complete P_2-differentiated kernel one-form displayed in the first line of (64). Rewriting the resulting logarithmic two-form in the original integral family gives [124]:

$$\begin{aligned} &I[\{0,0,0,0\},\{0,1\},\{0,0\}] + \frac{1}{2}\mathcal{C}_{\{0,1,0,1\}},\\ &I[\{0,0,1,1\},\{0,1\},\{0,0\}] - \frac{1}{2}\mathcal{C}_{\{0,1,1,0\}},\\ &I[\{1,1,1,1\},\{0,1\},\{0,0\}] + \frac{1}{2}\mathcal{C}_{\{1,0,1,0\}},\\ &\mathcal{C}_{\boldsymbol{n}} \equiv k_s^2 I[\{\boldsymbol{n}\},\{0,1\},\{1,1\}] - I[\{\boldsymbol{n}\},\{0,1\},\{-1,1\}]\\ &\qquad - I[\{\boldsymbol{n}\},\{0,1\},\{1,-1\}]. \end{aligned} \tag{28}$$

Here all integrals are evaluated at $d = 1-2\epsilon$.

(c) Replacing the Gram form in class (a) by $\mathrm{d}\log x_1$ gives the second scalar pairing: $\Phi_{\boldsymbol{mn}} \propto \delta_{\boldsymbol{mn}} \frac{\mathrm{d}x_1\,\mathrm{d}x_2}{x_1 x_2}$, and thus

$$\begin{aligned} &k_s I[\{0,0,0,0\},\{0,0\},\{2,2\},d=3-2\epsilon]\,,\\ &k_s I[\{0,0,1,1\},\{0,0\},\{2,2\},d=3-2\epsilon]\,,\\ &k_s I[\{1,1,1,1\},\{0,0\},\{2,2\},d=3-2\epsilon]\,. \end{aligned} \tag{29}$$

(d) Another logarithmic pairing is obtained by placing the P_2-connection along the x_1 fiber and pairing it with $\mathrm{d}\log x_2$, where we have $\Phi_{\boldsymbol{mn}} \propto (\partial_{P_2}\Omega_{ex})_{\boldsymbol{mn}}\mathrm{d}x_1 \frac{\mathrm{d}x_2}{x_2}$ and

$$\begin{aligned} &k_s I[\{0,1,0,0\},\{0,1\},\{1,2\},d=3-2\epsilon]\,,\\ &k_s I[\{0,1,1,1\},\{0,1\},\{1,2\},d=3-2\epsilon]\,,\\ &k_s I[\{1,0,0,0\},\{0,1\},\{1,2\},d=3-2\epsilon]\,,\\ &k_s I[\{1,0,1,1\},\{0,1\},\{1,2\},d=3-2\epsilon]\,. \end{aligned} \tag{30}$$

(e) Finally, the choice $\Phi_{\boldsymbol{mn}} \propto (\partial_{x_2}\Omega_{ex})_{\boldsymbol{mn}}\mathrm{d}x_1\,\mathrm{d}\log x_2$ requires an additional x_2-IBP. The IBP generates a separately logarithmic Gram term; after subtracting it, the two independent representatives in the original family are

$$\begin{aligned} &(2+b_0+2\nu)k_s I[\{0,1,0,1\},\{0,0\},\{1,3\},d=3-2\epsilon]\,,\\ &(2+b_0+2\nu)k_s I[\{0,1,1,0\},\{0,0\},\{1,3\},d=3-2\epsilon]\,. \end{aligned} \tag{31}$$

The IBP relation and the Gram subtraction are detailed in the Supplemental Material.

The alternative pairing of the Gram form in class (a) with $(\partial_{x_2}\Omega_{ex})\mathrm{d}x_2$ reduces to class (a) after the analogous IBP and Gram subtraction, and therefore gives no additional representative. After imposing the symmetry relations and parity constraints, the five classes yield $2+3+3+4+2 = 14$ independent representatives, in agreement with the top-sector IBP count for the even-even subsystem. The $d = 1-2\epsilon$ representatives in classes (a) and (b) can first be shifted to $d = 3-2\epsilon$ using the dimensional recurrence relation (67), after which the IBP reduction is performed uniformly.

The homogeneous 14×14 top-to-top block of the differential equations for the basis above is verified to be of $\mathrm{d}\log$-form. The corresponding alphabet is

$$\mathcal{A} = \{x-1, x, x+1\}, \tag{32}$$

with $x = k_s/P_1 = k_s/P_2$. The $\mathrm{d}\log$ decompositions of both this homogeneous block and the complete 19×19 connection, including the remaining-term family, are provided in the Supplemental Material. The three letters have clear physical intuition: x, $x+1$, and $x-1$ correspond to the soft limit $k_s = 0$, the partial-energy pole $P_{1,2}+k_s = 0$, and the spurious folded pole $P_{1,2}-k_s = 0$, respectively, while the total-energy pole $P_1+P_2 = 0$ corresponds to $x\to\infty$, where the differential system has its residue at infinity.

IV. CONCLUSION

In this Letter, we study IBP reduction and differential equations for the one-loop bubble family in dS space. We first identify the parity structure of the dS IBP system, which reduces the size of the IBP system for an n-propagator family by a factor of 2^n, and thus significantly enhances the feasibility of this computational approach. We successfully apply the IBP reduction and the method of differential equations to loop-level massive cosmological correlators. This demonstrates the viability of these methods as a systematic and automated framework for perturbative field theory calculations in dS space. We also explore the interesting mathematical structure of this system. We further propose that fibration intersection theory can be generalized to cases in which the twist $\mathbf{U}$ is not of polynomial type. Based on this, we conjecture that selecting a $\mathrm{d}\log$-form twist connection and choosing $\mathrm{d}\log$-form representatives Φ for

the master integrals can lead directly to d log-form differential equations. We adapt the Baikov representation to the dS case and find it useful for the construction of d log-form Φ. The results for the bubble example verify the conjecture and yield a simple alphabet. This structure may be useful for future analytic evaluations of the system.

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SUPPLEMENTARY MATERIAL

1. Reduction relations and kinematic differential operators for top-sector families

As mentioned in [80], for bubble integral families involving θ_{12} or θ_{21}, time IBP relations can collapse one of the time integrations and produce tadpole families. In this case, the collapsed propagator gives the following factor:

$$h^{(1)}_{n_1}(-k\tau_i)\,h^{(2)}_{n_2}(-k\tau_i) - h^{(2)}_{n_1}(-k\tau_i)\,h^{(1)}_{n_2}(-k\tau_i) = (n_1-n_2)e^{\pi\mathrm{Im}[\nu]}\frac{4\mathrm{i}}{\pi}(-k\tau_i)^{-2\nu-1}\,. \tag{33}$$

Note that with the extra factor $e^{\pi\mathrm{Im}[\nu]}$, the above formula applies to both real and imaginary ν. In our integral-family language, it is implemented as a shift of the corresponding $|\mathbf{k}|$-power index by $2\nu+1$ and the τ-power index by $-2\nu-1$, while the remaining families contain only one time integration. Such terms are known as remaining terms [80], and will be denoted by R.

First, let us consider the time-IBP relations. For ∂_{τ_1}, we have:

$$\begin{aligned}0 =& -\mathrm{i}P_1 I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_1 I[\{n_1,n_2,n_3,n_4\},\{a_1-1,a_2\},\{b_1,b_2\}]\\ &- I[\{n_1+1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - I[\{n_1,n_2,n_3+1,n_4\},\{a_1,a_2\},\{b_1,b_2-1\}]\\ &- (n_1-n_2)R_1[\{n_3,n_4\},\{a_1+a_2-2\nu-1\},\{b_1+2\nu+1,b_2\}]\\ &- (n_3-n_4)R_2[\{n_1,n_2\},\{a_1+a_2-2\nu-1\},\{b_1,b_2+2\nu+1\}]\,,\end{aligned} \tag{34}$$

and similarly for ∂_{τ_2}:

$$\begin{aligned}0 =& -\mathrm{i}P_2 I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_2 I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2-1\},\{b_1,b_2\}]\\ &- I[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - I[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2\},\{b_1,b_2-1\}]\\ &+ (n_1-n_2)R_1[\{n_3,n_4\},\{a_1+a_2-2\nu-1\},\{b_1+2\nu+1,b_2\}]\\ &+ (n_3-n_4)R_2[\{n_1,n_2\},\{a_1+a_2-2\nu-1\},\{b_1,b_2+2\nu+1\}]\,,\end{aligned} \tag{35}$$

where ν is the common mass parameter.

As for the momentum-IBP relations, we introduce

$$x_1 \equiv |\mathbf{q}|\,,\qquad x_2 \equiv |\mathbf{q}+\mathbf{k}_s|\,,\qquad \mathcal{O}_1 \equiv \mathbf{q}^i\partial_{\mathbf{q}^i}\,,\qquad \mathcal{O}_2 \equiv (\mathbf{q}^i+\mathbf{k}_s^i)\partial_{\mathbf{q}^i}\,, \tag{36}$$

whose actions on (x_1,x_2) are

$$\begin{aligned}&\mathcal{O}_1 x_1 = x_1\,,\quad \mathcal{O}_1 x_2 = \frac{1}{2}x_2^{-1}(x_1^2+x_2^2-k_s^2)\,,\\ &\mathcal{O}_2 x_2 = x_2\,,\quad \mathcal{O}_2 x_1 = \frac{1}{2}x_1^{-1}(x_1^2+x_2^2-k_s^2)\,.\end{aligned} \tag{37}$$

Equivalently, for any function $F(x_1,x_2)$, one has the chain rule

$$\mathcal{O}_\alpha F = (\mathcal{O}_\alpha x_1)\,\partial_{x_1}F + (\mathcal{O}_\alpha x_2)\,\partial_{x_2}F\,,\qquad \alpha=1,2\,. \tag{38}$$

From $\int \partial_{\mathbf{q}^i}[\mathbf{q}^i\cdots]=0$ one may write an intermediate step in terms of the composite operator acting on the full integrand,

$$\begin{aligned}0 =& \int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\int\frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\Big[d+\mathcal{O}_1\Big]\\ &\left[e^{-\mathrm{i}P_1\tau_1-\mathrm{i}P_2\tau_2}\frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1}x_2^{b_2}}\,h^{(1)}_{n_1}(-x_1\tau_1)h^{(2)}_{n_2}(-x_1\tau_2)\times h^{(1)}_{n_3}(-x_2\tau_1)h^{(2)}_{n_4}(-x_2\tau_2)\right]\\ =& \int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\int\frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\left[d+(\mathcal{O}_1x_1)\partial_{x_1}+(\mathcal{O}_1x_2)\partial_{x_2}\right]\\ &\left[e^{-\mathrm{i}P_1\tau_1-\mathrm{i}P_2\tau_2}\frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1}x_2^{b_2}}\,h^{(1)}_{n_1}(-x_1\tau_1)h^{(2)}_{n_2}(-x_1\tau_2)\times h^{(1)}_{n_3}(-x_2\tau_1)h^{(2)}_{n_4}(-x_2\tau_2)\right].\end{aligned} \tag{39}$$

Expanding the derivatives using $\partial_x x^{-b} = -(b/x)x^{-b}$ and $\partial_x h_n(-x\tau) = (-\tau)\, h_{n+1}(-x\tau)$ yields the closed relation in terms of shifted I integrals:

$$
\begin{aligned}
0 = {} & d\, I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - b_1\, I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
& - \frac{1}{2} b_2 \Big(I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] + I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-2,b_2+2\}] \\
& - k_s^2\, I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2+2\}] \Big) + I[\{n_1+1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1-1,b_2\}] \\
& + I[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2+1\},\{b_1-1,b_2\}] + \frac{1}{2} \Big(I[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2-1\}] \\
& + I[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1-2,b_2+1\}] - k_s^2\, I[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2+1\}] \Big) \\
& + \frac{1}{2} \Big(I[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2-1\}] + I[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1-2,b_2+1\}] \\
& - k_s^2\, I[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2+1\}] \Big) ,
\end{aligned} \tag{40}
$$

where d is the spatial dimension (in dimensional regularization $d = 3 - 2\epsilon$). The second momentum IBP can be written in the same chain-rule form as

$$
\begin{aligned}
0 = {} & \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^d \mathbf{q}}{(2\pi)^d} \Big[d + \mathcal{O}_2 \Big] \\
& \left[e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1} x_2^{b_2}}\, h^{(1)}_{n_1}(-x_1\tau_1) h^{(2)}_{n_2}(-x_1\tau_2) \times h^{(1)}_{n_3}(-x_2\tau_1) h^{(2)}_{n_4}(-x_2\tau_2) \right] ,
\end{aligned} \tag{41}
$$

which gives the symmetric relation with the exchanges

$$
\begin{aligned}
\{n_1,n_2,n_3,n_4\} &\leftrightarrow \{n_3,n_4,n_1,n_2\} , \\
\{b_1,b_2\} &\leftrightarrow \{b_2,b_1\} .
\end{aligned}
$$

Note that there is another set of relations following from (3), called EOM relations:

$$
\begin{aligned}
I[\{2,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = {} & - I[\{0,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
& - (2\nu+1) I[\{1,n_2,n_3,n_4\},\{a_1-1,a_2\},\{b_1+1,b_2\}] , \qquad (42) \\
I[\{n_1,2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = {} & - I[\{n_1,0,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
& - (2\nu+1) I[\{n_1,1,n_3,n_4\},\{a_1,a_2-1\},\{b_1+1,b_2\}] , \qquad (43) \\
I[\{n_1,n_2,2,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = {} & - I[\{n_1,n_2,0,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
& - (2\nu+1) I[\{n_1,n_2,1,n_4\},\{a_1-1,a_2\},\{b_1,b_2+1\}] , \qquad (44) \\
I[\{n_1,n_2,n_3,2\},\{a_1,a_2\},\{b_1,b_2\}] = {} & - I[\{n_1,n_2,n_3,0\},\{a_1,a_2\},\{b_1,b_2\}] \\
& - (2\nu+1) I[\{n_1,n_2,n_3,1\},\{a_1,a_2-1\},\{b_1,b_2+1\}] , \qquad (45)
\end{aligned}
$$

where ν is the common mass parameter.

In addition to the above IBP relations, we also have some symmetry relations:

$$
I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = I[\{n_3,n_4,n_1,n_2\},\{a_1,a_2\},\{b_2,b_1\}] . \tag{46}
$$

As our final step, we will present the differential equations of this integral family. Although we set $P_1 = P_2$, here we give the P_1 and P_2 derivatives separately:

$$
\frac{\partial}{\partial P_1} I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = \mathrm{i}\, I[\{n_1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1,b_2\}] , \tag{47}
$$

$$
\frac{\partial}{\partial P_2} I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = \mathrm{i}\, I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2+1\},\{b_1,b_2\}] . \tag{48}
$$

From

$$
\frac{\partial}{\partial k_s} |\mathbf{q}+\mathbf{k}_s| = \frac{k_s^2 + \mathbf{k}_s \cdot \mathbf{q}}{k_s |\mathbf{q}+\mathbf{k}_s|} = \frac{k_s^2 + |\mathbf{q}+\mathbf{k}_s|^2 - |\mathbf{q}|^2}{2 k_s |\mathbf{q}+\mathbf{k}_s|} , \tag{49}
$$

the k_s-derivative gives:

$$\begin{aligned}
&\frac{\partial}{\partial k_s} I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&= -\frac{b_2}{2k_s}\Big(k_s^2\, I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2+2\}] + I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&\quad - I[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-2,b_2+2\}]\Big) + \frac{1}{2k_s}\Big(k_s^2\, I[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2+1\}] \\
&\quad + I[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2-1\}] - I[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1-2,b_2+1\}]\Big) \\
&\quad + \frac{1}{2k_s}\Big(k_s^2\, I[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2+1\}] + I[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2-1\}] \\
&\quad - I[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1-2,b_2+1\}]\Big)\,.
\end{aligned} \tag{50}$$

The family also satisfies the scaling relation:

$$\begin{aligned}
&\left(k_s\frac{\partial}{\partial k_s} + P_1\frac{\partial}{\partial P_1} + P_2\frac{\partial}{\partial P_2}\right) I[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] \\
&= (d-b_1-b_2-a_1-a_2-2) I[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}]\,.
\end{aligned} \tag{51}$$

In conclusion, the IBP relations for the top sector are (34), (35), (40), and (41) together with the symmetry relations and the EOM relations. The differential equations for the top sector are (47), (48), and (50). The scaling relation (51) is used as a consistency check on the differential equations.

2. Reduction relations and kinematic differential operators for tadpole families R_1, R_2

In the IBP relations of the bubble diagram, tadpole integral families appear as remaining terms.

$$\begin{aligned}
R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] \equiv\; & e^{\pi \mathrm{Im}[\nu]}\frac{4\mathrm{i}}{\pi}\int_{-\infty}^{0}\mathrm{d}\tau\int\frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\,e^{-\mathrm{i}P_0\tau}\frac{(-\tau)^a}{|\mathbf{q}|^{b_1}|\mathbf{q}+\mathbf{k}_s|^{b_2}} \\
& \times\frac{1}{2}\sum_{r=0,1} h_{n_3}^{(1+r)}(-|\mathbf{q}+\mathbf{k}_s|\tau)h_{n_4}^{(2-r)}(-|\mathbf{q}+\mathbf{k}_s|\tau)\,,
\end{aligned} \tag{52}$$

$$\begin{aligned}
R_2[\{n_1,n_2\},\{a\},\{b_1,b_2\}] \equiv\; & e^{\pi \mathrm{Im}[\nu]}\frac{4\mathrm{i}}{\pi}\int_{-\infty}^{0}\mathrm{d}\tau\int\frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\,e^{-\mathrm{i}P_0\tau}\frac{(-\tau)^a}{|\mathbf{q}|^{b_1}|\mathbf{q}+\mathbf{k}_s|^{b_2}} \\
& \times\frac{1}{2}\sum_{r=0,1} h_{n_1}^{(1+r)}(-|\mathbf{q}|\tau)h_{n_2}^{(2-r)}(-|\mathbf{q}|\tau)\,.
\end{aligned} \tag{53}$$

Such terms have their own IBP relations. Here we only give the IBP relations for R_1, and the corresponding relations for R_2 follow via the symmetry $\mathbf{q}\to-(\mathbf{q}+\mathbf{k}_s)$, which exchanges $b_1\leftrightarrow b_2$ and relabels Hankel indices appropriately.

$$R_2[\{n_1,n_2\},\{a\},\{b_1,b_2\}] = R_1[\{n_1,n_2\},\{a\},\{b_2,b_1\}]. \tag{54}$$

The family R defined in the main text is R_2. Equivalently, using (54),

$$R[\{\boldsymbol{n}\},\{a\},\{b_1,b_2\}] = R_2[\{\boldsymbol{n}\},\{a\},\{b_1,b_2\}] = R_1[\{\boldsymbol{n}\},\{a\},\{b_2,b_1\}].$$

We therefore give the reduction relations only for R_1. The time IBP relation for R_1 is

$$\begin{aligned}
0 =& -\mathrm{i}(P_1+P_2)R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] - aR_1[\{n_3,n_4\},\{a-1\},\{b_1,b_2\}] \\
& - R_1[\{n_3+1,n_4\},\{a\},\{b_1,b_2-1\}] - R_1[\{n_3,n_4+1\},\{a\},\{b_1,b_2-1\}]\,,
\end{aligned} \tag{55}$$

and two momentum IBP relations read

$$\begin{aligned}0 =& dR_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] - b_1 R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}]\\ &- \frac{1}{2}b_2\Big(R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] + R_1[\{n_3,n_4\},\{a\},\{b_1-2,b_2+2\}] - k_s^2 R_1[\{n_3,n_4\},\{a\},\{b_1,b_2+2\}]\Big)\\ &+ \frac{1}{2}\Big(R_1[\{n_3+1,n_4\},\{a+1\},\{b_1,b_2-1\}] + R_1[\{n_3+1,n_4\},\{a+1\},\{b_1-2,b_2+1\}]\\ &- k_s^2 R_1[\{n_3+1,n_4\},\{a+1\},\{b_1,b_2+1\}]\Big) + \frac{1}{2}\Big(R_1[\{n_3,n_4+1\},\{a+1\},\{b_1,b_2-1\}]\\ &+ R_1[\{n_3,n_4+1\},\{a+1\},\{b_1-2,b_2+1\}] - k_s^2 R_1[\{n_3,n_4+1\},\{a+1\},\{b_1,b_2+1\}]\Big)\,. \end{aligned} \tag{56}$$

$$\begin{aligned}0 =& dR_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] - b_2 R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}]\\ &- \frac{1}{2}b_1\Big(R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] + R_1[\{n_3,n_4\},\{a\},\{b_1+2,b_2-2\}]\\ &- k_s^2 R_1[\{n_3,n_4\},\{a\},\{b_1+2,b_2\}]\Big) + R_1[\{n_3+1,n_4\},\{a+1\},\{b_1,b_2-1\}]\\ &+ R_1[\{n_3,n_4+1\},\{a+1\},\{b_1,b_2-1\}]\,. \end{aligned} \tag{57}$$

These two relations come from $\int \partial_{\mathbf{q}^i}[\mathbf{q}^i\cdots] = 0$ and $\int \partial_{\mathbf{q}^i}[(\mathbf{q}^i+\mathbf{k}_s^i)\cdots] = 0$, respectively.

Note that in the tadpole case, we also have relations coming from (3):

$$\begin{aligned}R_1[\{2,n_4\},\{a\},\{b_1,b_2\}] =& - R_1[\{0,n_4\},\{a\},\{b_1,b_2\}]\\ &- (2\nu+1)R_1[\{1,n_4\},\{a-1\},\{b_1,b_2+1\}]\,, \end{aligned} \tag{58}$$

$$\begin{aligned}R_1[\{n_3,2\},\{a\},\{b_1,b_2\}] =& - R_1[\{n_3,0\},\{a\},\{b_1,b_2\}]\\ &- (2\nu+1)R_1[\{n_3,1\},\{a-1\},\{b_1,b_2+1\}]\,. \end{aligned} \tag{59}$$

and the following symmetry relation

$$R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] = R_1[\{n_4,n_3\},\{a\},\{b_1,b_2\}]\,. \tag{60}$$

Let us turn to the differential equations. Taking $P_1 = P_2 = P_0/2$, the P_0-derivative gives:

$$\frac{\partial}{\partial P_0}R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] = \mathrm{i}R_1[\{n_3,n_4\},\{a+1\},\{b_1,b_2\}]\,. \tag{61}$$

and the k_s-derivative gives

$$\begin{aligned}&\frac{\partial}{\partial k_s}R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}]\\ =& - \frac{b_2}{2k_s}\Big(k_s^2\,R_1[\{n_3,n_4\},\{a\},\{b_1,b_2+2\}] + R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] - R_1[\{n_3,n_4\},\{a\},\{b_1-2,b_2+2\}]\Big)\\ &+ \frac{1}{2k_s}\Big(k_s^2\,R_1[\{n_3+1,n_4\},\{a+1\},\{b_1,b_2+1\}] + R_1[\{n_3+1,n_4\},\{a+1\},\{b_1,b_2-1\}]\\ &- R_1[\{n_3+1,n_4\},\{a+1\},\{b_1-2,b_2+1\}]\Big) + \frac{1}{2k_s}\Big(k_s^2\,R_1[\{n_3,n_4+1\},\{a+1\},\{b_1,b_2+1\}]\\ &+ R_1[\{n_3,n_4+1\},\{a+1\},\{b_1,b_2-1\}] - R_1[\{n_3,n_4+1\},\{a+1\},\{b_1-2,b_2+1\}]\Big)\,. \end{aligned} \tag{62}$$

Similarly to the bubble case, we have the following scaling relation:

$$\left(k_s\frac{\partial}{\partial k_s} + P_0\frac{\partial}{\partial P_0}\right)R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] = (d-b_1-b_2-a-1)R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}]\,, \tag{63}$$

which will be used to verify the correctness of the differential equations.

In conclusion, the IBP relations for the remaining terms are (55), (56), and (57) together with symmetry relations and EOM relations. The differential equations are (61) and (62). As before, the remaining terms also satisfy the scaling relation (63).

3. Details for $\mathrm{d}\log$-form building blocks involving Ω^τ

The first expression below gives the logarithmic one-form used in class (b), while the remaining identities justify those used in classes (d) and (e). Since Ω_{ex} contains only $\log(P_i \pm x_1 \pm x_2)$, the required combinations can be written as

$$\begin{aligned}\mathrm{d}\Big[\int (\partial_{P_2}\Omega_{ex})_{\boldsymbol{mn}}\mathrm{d}x_2\Big] &= \left[\Big[(-1)^{\lfloor|\boldsymbol{m}|\rfloor/2}\frac{P_2^2-x_1^2-x_2^2}{2x_1x_2}\partial_{P_2}\Omega_{ex}\Big]_{\boldsymbol{mn}} + (\Omega_{re})_{\boldsymbol{mn}}\right]\mathrm{d}x_1 + (\partial_{P_2}\Omega_{ex})_{\boldsymbol{mn}}\,\mathrm{d}x_2,\\ \partial_{x_j}\Omega_{ex}\,\mathrm{d}x_i &= \left(\partial_{x_j}\Omega^\tau - \partial_{x_j}\Omega_{x_j}\right)\mathrm{d}x_i\,.\end{aligned} \tag{64}$$

where Ω_{re} is given in the supplementary files. Here and below, $\hat{I}$ denotes the corresponding Baikov form before the x_1 and x_2 integrations. Using

$$\begin{aligned}\hat{I}[\{\boldsymbol{n}\},\{a_1+1,a_2\},\{b_1,b_2\}] &= -\mathrm{i}\partial_{P_1}\hat{I}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] = -\mathrm{i}\,(\partial_{P_1}\Omega^\tau)_{\boldsymbol{nm}}\,\hat{I}[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1,b_2\}]\\ &= -\mathrm{i}\,(\partial_{P_1}\Omega_{ex})_{\boldsymbol{nm}}\,\hat{I}[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1,b_2\}]\\ (b_0+1)\hat{I}[\{\boldsymbol{n}\},\{0,0\},\{1,3\}] &= (\partial_{x_2}\Omega^\tau)_{\boldsymbol{nm}}\,\hat{I}[\{\boldsymbol{m}\},\{0,0\},\{1,2\}] - \epsilon\partial_{x_2}\log\mathcal{G}\hat{I}[\{\boldsymbol{n}\},\{0,0\},\{1,2\}],\\ &= (\partial_{x_2}\Omega_{ex}+\partial_{x_2}\Omega_{x_2})_{\boldsymbol{nm}}\,\hat{I}[\{\boldsymbol{m}\},\{0,0\},\{1,2\}] - \epsilon\frac{\partial_{x_2}\mathcal{G}}{\mathcal{G}x_2}\hat{I}[\{\boldsymbol{n}\},\{0,0\},\{1,1\}]\end{aligned} \tag{65}$$

and notice that the $\partial\mathcal{G}$ term is also a $\mathrm{d}\log$-form

$$\frac{\partial_{x_2}\mathcal{G}\mathrm{d}x_1\mathrm{d}x_2}{\mathcal{G}x_2} = 2\left[\mathrm{d}\tanh^{-1}\left(\frac{x_1}{k_s+x_2}\right) - \mathrm{d}\tanh^{-1}\left(\frac{x_1}{k_s-x_2}\right)\right]\mathrm{d}\log x_2\,.$$

One finds $\partial_{P_j}\Omega_{ex}\,\mathrm{d}x_i \propto \tau_j\,\mathrm{d}x_i$ and $\partial_{x_j}\Omega_{ex}\,\mathrm{d}x_i\;\mathrm{d}\log x_j + (\partial\mathcal{G}\text{ term}) \propto \frac{1}{x_j^2}\mathrm{d}x_i\mathrm{d}x_j$ as additional $\mathrm{d}\log$-form building blocks. Here $\propto$ is understood up to IBP equivalence.

4. Raising $d=1$ back to $d=3$

In this section, we demonstrate that all dimension-1 master integrals can be raised to their corresponding dimension-3 master integrals. Therefore, they do not require special treatment in the differential equations.

For the normalized Baikov measure, one has

$$\frac{2}{\pi^{d/2}}\,\mathrm{d}^d\mathbf{q} = C_d\,\mathcal{G}^{\frac{d-3}{2}}\mathcal{K}^{-\frac{d-2}{2}}\,\mathrm{d}z_1\,\mathrm{d}z_2\,. \tag{66}$$

In particular, for $d=1-2\epsilon$ and $d=3-2\epsilon$,

$$C_{1-2\epsilon}\,\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2} = \partial_{q^2}\Big[C_{3-2\epsilon}\,\mathcal{K}^{-1/2+\epsilon}\mathcal{G}^{-\epsilon}\Big]\,, \tag{67}$$

where $\partial_{q^2}\equiv\partial_{z_1}+\partial_{z_2}$ and we used $\partial_{q^2}\mathcal{G}=\mathcal{K}$ together with $\Gamma(1-\epsilon)=-\epsilon\,\Gamma(-\epsilon)$.

Therefore, for

$$\begin{aligned}\hat{I}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] &= \mathrm{d}z_1\,\mathrm{d}z_2\,z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2}\,I^\tau_{\boldsymbol{n}}\\ &\quad\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}\,C_{1-2\epsilon}\,\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2}\,,\end{aligned} \tag{68}$$

Integration by parts gives

$$\begin{aligned}\hat{I}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] &= \mathrm{d}z_1\,\mathrm{d}z_2\left[\left(\frac{b_0+b_1}{2z_1}+\frac{b_0+b_2}{2z_2}\right)I^\tau_{\boldsymbol{n}} - (\partial_{z_1}+\partial_{z_2})\,I^\tau_{\boldsymbol{n}}\right]\\ &\quad\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}\,z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2}\,C_{3-2\epsilon}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}\,.\end{aligned} \tag{69}$$

Equivalently, in (x_1,x_2) variables with $\partial_{q^2}=\frac{1}{2x_1}\partial_{x_1}+\frac{1}{2x_2}\partial_{x_2}$,

$$\begin{aligned}\hat{I}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] &= 2\,\mathrm{d}x_1\,\mathrm{d}x_2\left[\left(\frac{b_0+b_1}{x_1^2}+\frac{b_0+b_2}{x_2^2}\right)I^\tau_{\boldsymbol{n}} - \left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right)I^\tau_{\boldsymbol{n}}\right]\\ &\quad\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}\,x_1^{-(b_0+b_1-1)}x_2^{-(b_0+b_2-1)}\,C_{3-2\epsilon}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}\,.\end{aligned} \tag{70}$$

In particular, for the $d=1$ candidates above,

$$\begin{aligned}
\hat{I}[\{\boldsymbol{n}\},\{0,0\},\{1,0\};d=1] &= \mathrm{d}z_1\,\mathrm{d}z_2\, z_1^{-b_0/2-1/2} z_2^{-b_0/2}\, I^{\tau}_{\boldsymbol{n}}\, \mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2} \\
&= 2\,\mathrm{d}x_1\,\mathrm{d}x_2 \left[\left(\frac{b_0+1}{x_1^2}+\frac{b_0}{x_2^2}\right) I^{\tau}_{\boldsymbol{n}} - \left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right) I^{\tau}_{\boldsymbol{n}}\right] \\
&\quad \times x_1^{-b_0} x_2^{-b_0+1}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}, \\
\hat{I}[\{\boldsymbol{n}\},\{1,0\},\{0,0\};d=1] &= 2\,\mathrm{d}x_1\,\mathrm{d}x_2 \left[b_0\left(\frac{1}{x_1^2}+\frac{1}{x_2^2}\right) I^{\tau}_{\boldsymbol{n}} - \left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right) I^{\tau}_{\boldsymbol{n}}\right] \\
&\quad \times (-\tau_1)\, x_1^{-b_0+1} x_2^{-b_0+1}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}\,. \qquad (71)
\end{aligned}$$

Hence, all $d=1$ master integrals can be rewritten as $d=3$ master integrals.